



HORMONES, SEXUAL HEALTH & MISC

Melanotan-1

Melanotan-1 – 5 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	Afamelanotide ([Nle4, D-Phe7]-alpha-MSH)
SYNONYMS / ALIASES	Melanotan-1; MT-1; afamelanotide; NDP-MSH
CAS NUMBER	75921-69-6
MOLECULAR FORMULA	C78H111N21O19
MOLECULAR WEIGHT	1646.85 g/mol
AMINO ACID SEQUENCE	Ac-Ser-Tyr-Ser-Nle-Glu-His-D-Phe-Arg-Trp-Gly-Lys-Pro-Val-NH2
SOURCE	Synthetic (solid-phase peptide synthesis)

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized white to off-white powder
PURITY SPECIFICATION	>=99% by HPLC (confirm against batch COA)
CONTAINER	5 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	-20 C, protect from light and moisture
RECONSTITUTED STORAGE	2-8 C, stable approximately 14-28 days in bacteriostatic water
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only)
SHELF LIFE (UNRECONSTITUTED, SEALED)	24 months at -20 C
STABILITY NOTES	Avoid repeated freeze-thaw cycles.

STRUCTURAL NOTES

Melanotan-1 is a linear synthetic analog of alpha-melanocyte-stimulating hormone (alpha-MSH) with norleucine at position 4 and D-phenylalanine at position 7 conferring metabolic stability. The cyclic analog is described in the [[melanotan-ii]] datasheet. This entry provides a descriptive structural classification only.

SELECTED PEER-REVIEWED REFERENCES

1. Sawyer T.K. et al. - *4-Norleucine, 7-D-phenylalanine-alpha-melanocyte-stimulating hormone: a highly potent alpha-melanotropin with ultralong biological activity* - Proc Natl Acad Sci USA, 1980 - DOI: 10.1073/pnas.77.10.5754
2. Langan E.A. et al. - *Melanotropic peptides* - (review) [VERIFY WITH COA]

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

COMPLIANCE DISCLAIMER

This product is sold by Aligned Aminos for laboratory research use only (RUO). It is not a drug, dietary supplement, food, cosmetic, or medical device. It has not been approved by the FDA for any therapeutic, diagnostic, or preventative use. It is not intended for use in humans or animals, and may not be administered, ingested, applied, or otherwise introduced into any living organism. Purchasers represent that they are qualified researchers or laboratory professionals and assume full responsibility for compliance with all applicable laws and institutional review requirements governing the handling of research chemicals.

Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.